Lesson 02 — Storage Devices, Direct Data Entry, Printers and File Management

Strand 1 — Introduction to Computing · Sub-strand 1 — Components of Computers and Computer Systems Indicators: B7.1.1.2.2 (file & folder management) · B8.1.1.1.2 (direct data entry devices) · B8.1.1.1.3 (printers) · B8.1.1.1.4 (storage devices) · B8.1.1.2.2 (drive management) · B9.1.1.1.4 (network/cloud storage & smart cards) · B9.1.1.2.2 (file management techniques) Source coverage: BECE Mock 0201026/27 Computing (2026) P1 Q1, 6, 10, 26, 30, 34–37 · P2 Q1(a), 4(c), 5(c) + authentic bank questions.

Preamble (why this lesson first, in everyday words)

Think of a computer like a person. Its brain (CPU) thinks quickly, but a person also has a memory that keeps things for later. A computer's memory for later is called storage. Without storage, everything you type would vanish the moment the power goes off — like writing on water.

But before the computer can even store something, the information must enter the computer. Some information goes in through your fingers (typing). Other information goes in directly from the object itself — like a shop card with black and white lines that the computer "reads" by itself. That family of devices is called direct data entry devices.

And once information is inside, it must be tidily kept — that is why we group files into folders, just like exercise books are grouped into a school satchel.

Finally, information often needs to leave the computer on paper — and there are many kinds of printers, each choosing a different way to put marks on paper.

In this lesson we learn these four families: storage, direct data entry, printers, and file/folder & drive management. Everything we learn later — spreadsheets, word processing, networks — stands on this ground.

Words to know first

Word	Simple meaning
Storage	Where a computer keeps data permanently, even when switched off.
Capacity	How much data a device can hold, measured in megabytes (MB), gigabytes (GB) or terabytes (TB).
Drive	A storage device seen by the computer as a letter, e.g. C:, D:.
Magnetic	Uses spinning, magnet-coated disks to store data.
Optical	Uses a laser to read data from a shiny disk (CD/DVD).
Flash memory	Stores data on microchips with no moving parts.
Barcode	A series of parallel black and white lines that a machine reads as a number.
RFID	Radio Frequency Identification — tiny tags read by radio waves without touching.
Direct data entry (DDE)	Devices that read data straight from its source (card, document, badge) without typing.
Partition	A section of one hard disk that behaves like a separate drive.
Impact printer	A printer that strikes paper through an inked ribbon, like a typewriter.
Tactile output	Output you can read with your fingertips.

Teaching body (Theory Questions and Answers — Section 1: File and folder management)

Q1. What is a file? Ans. A file is a named collection of related data stored on a computer, such as a document, a picture or a song.

Q2. What is a folder? Ans. A folder is a container that stores and organises related files together.

Q3. Where are files stored? Ans. Files are stored in a folder.

Q4. Why do we organise files into folders? Ans. We organise files into folders so that related data is easy to find and we avoid losing or duplicating files.

Q5. Give the steps to create a new folder. Ans.

1. Right-click on an empty space on the desktop or inside the parent folder.
2. Hover over New.
3. Click Folder.
4. Type the folder name and press Enter.

Q6. State the steps to move a file into a folder. Ans. Right-click the file → click Cut → open the destination folder → right-click inside it → click Paste. (Alternatively drag the file onto the folder.)

Q7. What is a storage drive? Ans. A storage drive is a hardware device where the computer permanently keeps files and programs.

Q8. What is partitioning of a hard disk? Ans. Partitioning is dividing one physical hard disk into two or more separate logical drives, such as Drive C: and Drive D:.

Q9. Explain the purpose of creating multiple partitions on a hard disk. Ans. (Any two)

- To organise data separately, e.g. schoolwork on one drive and games on another.
- To separate the operating system from personal files, so reinstalling the system does not destroy personal data.
- To allow more than one operating system to be installed on the same computer.
- To make backup easier by copying one partition only.

(Real exam voice — Mock 0201026/27 P2 Q4(c): the student divides his hard disk into two visible drives.)

Q10. What is the Recycle Bin? Ans. The Recycle Bin is a special folder that temporarily keeps deleted files so they can be restored before being permanently removed.

Q11. What is a backup? Ans. A backup is a second copy of important files kept on a different storage device in case the original files are lost or damaged.

Teaching body — Section 2: Storage devices

Q12. What is a Hard Disk Drive (HDD)? Ans. An HDD is a magnetic storage device that records data on spinning platters using a moving read/write head.

Q13. What is a Solid-State Drive (SSD)? Ans. An SSD is a storage device that stores data on flash memory chips using no moving parts at all.

Q14. Why is an SSD described as robust and resistant to physical shocks? Ans. Because an SSD has no moving parts, a knock or drop cannot damage any mechanical part that would destroy the data.

Q15. Give two reasons an expert may recommend a Solid-State Drive over an HDD. Ans. (Any two)

- It has no moving parts, so it is more durable and shock-resistant.
- It reads and writes data much faster, so files open quickly.
- It is silent and produces no vibration.
- It uses less power, so it suits laptops.

(Real exam voice — Mock 0201026/27 P2 Q1(a)(iii).)

Q16. Give two differences between an HDD and an SSD.

Feature	HDD	SSD
Parts	Spinning platters + moving head	Flash memory chips only
Speed	Slower	Much faster
Durability	Sensitive to shocks	Shock-resistant
Price per GB	Cheaper	More expensive

Q17. What storage device is most likely used inside a smartphone? Ans. Embedded flash memory (an embedded SSD-type chip soldered inside the phone).

Q18. What is a flash drive (pen drive)? Ans. A flash drive is a small portable USB device that stores files on flash memory so they can be carried between computers.

Q19. What is an optical storage device? Ans. An optical device, such as a CD or DVD drive, uses a laser to read and write data on a shiny disk.

Q20. What is magnetic tape used for? Ans. Magnetic tape is used mainly for backing up very large amounts of data, because it is cheap per gigabyte.

Q21. What is cloud storage? Ans. Cloud storage keeps files on remote servers that you reach through the Internet, such as Google Drive or OneDrive.

Q22. Give two advantages of cloud storage. Ans. (Any two) Files can be reached from any device with Internet access; files are safe even if your device is stolen; it is easy to share files.

Q23. What is a smart card? Ans. A smart card is a plastic card with an embedded chip that stores data, such as an ID card or an ATM card.

Teaching body — Section 3: Direct data entry devices

Q24. What is a direct data entry device? Ans. A direct data entry device reads data directly from a source document or object and sends it straight into the computer without a person typing.

Q25. Name the feature of an ID card that shows black and white lines readable directly into the computer. Ans. A barcode.

(Real exam voice — Mock 0201026/27 P2 Q1(a)(i).)

Q26. Which device captures information from a barcode? Ans. A barcode reader (also called a barcode scanner).

(Real exam voice — Mock 0201026/27 P2 Q1(a)(ii).)

Q27. What does RFID stand for? Ans. Radio Frequency Identification.

Q28. How is RFID different from a barcode? Ans. An RFID tag is read by radio waves from a distance without line of sight, while a barcode must be scanned directly and can be read only by light.

Q29. Name three other direct data entry devices. Ans. Magnetic Ink Character Recognition (MICR) reader — reads bank cheque numbers; Optical Mark Recognition (OMR) — marks exam answer sheets; Optical Character Recognition (OCR) — reads typed or printed text from paper.

Q30. What is a magnetic card reader? Ans. A magnetic card reader is a device that reads data stored on the magnetic stripe of a card, such as when a card is swiped.

Teaching body — Section 4: Printers

Q31. What is a printer? Ans. A printer is an output device that produces a paper copy (hard copy) of information from the computer.

Q32. Name four types of printer. Ans. Impact (dot matrix), inkjet, thermal, laser, Braille, wax and 3D printers. (Any four.)

Q33. What is an impact printer? Ans. An impact printer produces characters by striking pins through an inked ribbon onto the paper.

Q34. Why choose an impact printer for multi-part carbon receipt forms? Ans. Because the striking force presses through several carbon copy sheets at once, so all copies receive the print — no other printer can do this.

Q35. What is the main difference between an impact printer and a thermal printer? Ans. An impact printer strikes the paper through an inked ribbon with pins, which is noisy and noisy-copying, while a thermal printer prints silently and instantly by applying heat to special heat-sensitive paper.

(Real exam voice — Mock 0201026/27 P2 Q5(c).)

Q36. Why is a thermal printer good for till receipts? Ans. It prints very fast, quietly and without ink, so short receipts come out instantly at the sales desk.

Q37. What is a Braille printer? Ans. A Braille printer is an impact printer that emboses raised dots on paper so that a visually impaired person can read the output with their fingertips.

Q38. Why does a Braille printer suit a storybook for visually impaired children? Ans. Because it can produce tactile output — words the children read by touch, not by sight.

(Real exam voice — Mock 0201026/27 P1 Q10.)

Q39. What is a 3D printer? Ans. A 3D printer builds a physical, three-dimensional solid object layer by layer from a digital model.

Q40. Why is a 3D printer the most suitable for producing an architectural building model? Ans. Because it turns the design file into a real three-dimensional physical model, which flat paper printers cannot do.

(Real exam voice — Mock 0201026/27 P1 Q35.)

Acronym table (memorise these)

Acronym	Full meaning
HDD	Hard Disk Drive
SSD	Solid-State Drive
USB	Universal Serial Bus
CD / DVD	Compact Disc / Digital Versatile Disc
RFID	Radio Frequency Identification
MICR	Magnetic Ink Character Recognition
OMR	Optical Mark Recognition
OCR	Optical Character Recognition
GB / TB	Gigabyte / Terabyte

Worked Exam Exemplars — 15 solved examples

Each example shows the exact question, the marks, and the model marking-scheme answer. Examples marked [Mock 0201026/27] come straight from your paper.

Example 1 [Mock 0201026/27, P1 Q1] A group of files is stored in a: graphic / folder / text / word. (1 mark) Ans. B — folder. A folder is the container that keeps related files together.

Example 2 [Mock 0201026/27, P1 Q26] Which storage device is known for being robust and resistant to physical shocks? — HDD / SSD / Optical Drive / Magnetic Tape. (1 mark) Ans. B — SSD. No moving parts means a drop or knock does not damage a spinning platter.

Example 3 [Mock 0201026/27, P1 Q30] Which storage device is most likely used in a smartphone? — HDD / SSD / Embedded Flash Memory / Optical Drive. (1 mark) Ans. C — Embedded Flash Memory. A phone is too thin and too light for a spinning hard disk; the storage chip is built (embedded) inside the phone.

Example 4 [Mock 0201026/27, P2 Q1(a)(i)] A unique student ID card has the student's name, card number, expiry date, and a series of black and white lines along the bottom edge read directly into the computer. Name the ID card feature that makes direct data entry possible. (2 marks) Ans. The barcode. (2 marks — naming "barcode"; saying it is machine-readable so no typing is needed.)

Example 5 [Mock 0201026/27, P2 Q1(a)(ii)] Which device could be used to capture student information from this feature? (2 marks) Ans. A barcode reader/scanner. Held over the card, the reader converts the black and white lines into the student's data directly in the computer.

Example 6 [Mock 0201026/27, P2 Q1(a)(iii)] An expert recommended a Solid-State Drive to store scanned student records. Give two reasons. (4 marks) Ans. (Any two, 2 marks each)

- No moving parts → durable and resistant to shocks, so valuable school records are safe.
- Very fast read/write speed → attendance data is saved and opened quickly every morning.
- Also accepted: silent operation; lower power use; no fragmentation.

Example 7 [Mock 0201026/27, P2 Q4(c)] A student divides his hard disk into two visible drives to keep schoolwork and games separate. Explain the purpose of creating multiple partitions on a hard disk. (2 marks) Ans. Partitioning splits one physical disk into independent drives so different kinds of data can be organised separately, protected separately, or backed up and reinstalled without touching the other partition.

Example 8 [Mock 0201026/27, P2 Q5(c)] The office prints multi-part carbon receipt forms, and the reception prints short till slips instantly and quietly. State the main difference between an impact printer and a thermal printer. (3 marks) Ans. An impact printer strikes pins through an inked ribbon onto paper to form characters and can print through carbon copies, whereas a thermal printer heats special heat-sensitive paper to print silently and instantly; thermal printers need special paper while impact printers use ordinary multi-part forms.

Example 9 [Mock 0201026/27, P1 Q10] A storybook for visually impaired children must be read with the fingertips. Which printer produces tactile output? — Inkjet / Thermal / Braille / 3D. (1 mark) Ans. C — Braille printer. It embosses raised dots readable by touch.

Example 10 [Mock 0201026/27, P1 Q35] An architecture student wants a physical 3D model of her design. Which printer is most suitable? — Impact / Inkjet / 3D / Thermal. (1 mark) Ans. C — 3D printer. It builds a solid object layer by layer from the digital model.

Example 11 [Own, Paper 2 style] State two differences between flash memory (SSD type) storage and magnetic (HDD type) storage. (2 marks) Ans. (1 mark each)

- Flash storage uses chips with no moving parts; magnetic storage spins platters with a moving head.
- Flash is faster and shock-resistant; magnetic is slower but cheaper per gigabyte.

Example 12 [Own, Paper 2 style] Mention three direct data entry devices used in Ghanaian offices or exam halls. (3 marks) Ans. (1 mark each) Barcode reader (shops and ID cards); OMR scanner (BECE answer sheets); MICR reader (bank cheques). Also RFID reader.

Example 13 [Own, Paper 2 style] Give two advantages of cloud storage such as Google Drive. (2 marks) Ans. (1 mark each) Files are reachable from any device with Internet access; files survive if your device is lost or damaged. Also easy sharing.

Example 14 [Own, Paper 1 converted to written answer] What is the difference between input and output in computing? Give one device for each. Ans. Input puts data into the computer (e.g. a barcode reader); output sends processed data out to people (e.g. a printer).

Example 15 [Own, Paper 2 style] A trader uses a till system that reads her customers' ID cards at once. State two reasons why direct data entry is better than typing the details by hand. (2 marks) Ans. (1 mark each) It is much faster; it removes human typing errors, so the captured records are accurate.

Exercises — 5 sections × 5 questions (no multiple choice; write full answers)

Instructions: answer in your exercise book in full sentences. Marks are shown in brackets, as BECE Paper 2 does.

Exercise 1 — Storage devices

1. What is storage in a computer system? (2 marks)
2. State two differences between a Hard Disk Drive and a Solid-State Drive. (2 marks)
3. Why is a Solid-State Drive recommended for storing important school records? (2 marks)
4. Name the storage device found inside a smartphone. (1 mark)
5. Which storage device uses a laser to read data? (1 mark)

Exercise 2 — Cloud, backup and partitions

1. What is cloud storage? (2 marks)
2. State two advantages of cloud storage. (2 marks)
3. What is partitioning of a hard disk? (2 marks)
4. Give two purposes of partitioning a school computer's hard disk. (2 marks)
5. Define a backup. (2 marks)

Exercise 3 — Direct data entry

1. What is a direct data entry device? (2 marks)
2. Name the feature on an ID card made of black and white lines that is read directly into the computer. (2 marks)
3. Name the device that captures this feature. (1 mark)
4. Write the full meaning of RFID. (1 mark)
5. Mention three other direct data entry devices. (3 marks)

Exercise 4 — Printers

1. What is a printer? (2 marks)
2. State the main difference between an impact printer and a thermal printer. (3 marks)
3. Why is an impact printer the only suitable choice for printing multi-part carbon receipt forms? (2 marks)
4. Name the printer that produces raised dots readable by the fingertips. (1 mark)
5. A school wants a solid model of a water tower for a science fair. Name the most suitable printer and the type of output it makes. (2 marks)

Exercise 5 — File, folder and drive management + review

1. Define a file and a folder. (2 marks)
2. Outline the steps for creating a new folder on the desktop. (4 marks)
3. What is a drive letter? Give one example. (2 marks)
4. [Review] A friend deleted an important document by mistake, but the file is still on the screen list of deleted items. Name the feature that can give it back. (2 marks)
5. [Review] State two reasons why a school should back up student attendance data on a second device. (2 marks)

Answers / marking scheme

Exercise 1

1. Storage is where a computer keeps data and programs permanently, so they survive after the computer switches off. (2 marks — "keeps data" 1, "permanent/off" 1)

2. HDD uses spinning platters and a moving head and is slower; SSD uses flash chips with no moving parts, is faster and shock-resistant. (1 mark each)
3. It has no moving parts, so shocks cannot destroy the data, and it stores files quickly and reliably. (2 marks)
4. Embedded flash memory. (1 mark)
5. An optical drive (CD/DVD). (1 mark)

Exercise 2

1. Cloud storage keeps files on remote servers that the user reaches through the Internet. (2 marks)
2. Any two: access from any internet-connected device; safe from loss/theft of the device; easy sharing. (1 mark each)
3. Partitioning divides one physical hard disk into two or more separate drives. (2 marks)
4. Any two: organise schoolwork and games separately; separate system files from personal data; easier backup; allow a second operating system. (1 mark each)
5. A backup is a second copy of data kept on a different storage device in case the original is lost or damaged. (2 marks)

Exercise 3

1. A direct data entry device reads data straight from a source object or document into the computer without typing. (2 marks)
2. The barcode. (2 marks)
3. A barcode reader (barcode scanner). (1 mark)
4. Radio Frequency Identification. (1 mark)
5. Any three: MICR reader; OMR scanner; OCR scanner; magnetic card reader; RFID reader. (1 mark each)

Exercise 4

1. A printer is an output device that produces a paper (hard) copy of computer information. (2 marks)
2. An impact printer strikes pins through an inked ribbon onto the paper (noisy, can print through carbon copies); a thermal printer uses heat on heat-sensitive paper (silent, instant). (3 marks — impact 1, thermal 1, both sides stated 1)
3. Its striking force presses through several carbon sheets at once, printing every copy together. (2 marks)
4. A Braille printer. (1 mark)
5. A 3D printer; it produces a solid three-dimensional physical model built layer by layer. (2 marks)

Exercise 5

1. A file is a named collection of related data (document, picture, song); a folder is a container that stores and organises related files. (1 mark each)
2. Right-click an empty space → hover on New → click Folder → type the name → press Enter. (1 mark each point)
3. The letter the computer uses to name a storage drive, e.g. C: or D:. (2 marks)
4. The Recycle Bin — deleted files wait there and can be restored. (2 marks)
5. Any two: if one device fails, attendance data is not lost; the backup allows records to be restored after a breakage or thief. (1 mark each)

Next lesson: Technology in the Community — assistive and adaptive technology, portable devices, and the computer's effect on the environment (this paper tests it heavily in P1 Q7, 8, 13 and P2 Q1(c)).